

Thermodynamics

Lecture 2C:

- Energy Transfer, General Energy Analysis
- Pure substances

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After this lecture

- Understand energy analysis of open systems and cycles.
- Define energy conversion efficiency.
- Introduction to pure substances and phase change processes.

Learning objectives

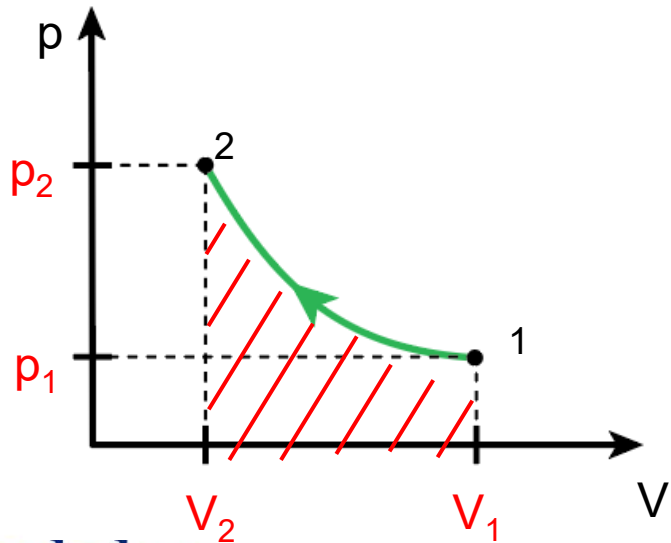
What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

Recap Lecture 2B

Boundary work

$$W = \int_1^2 p dV$$



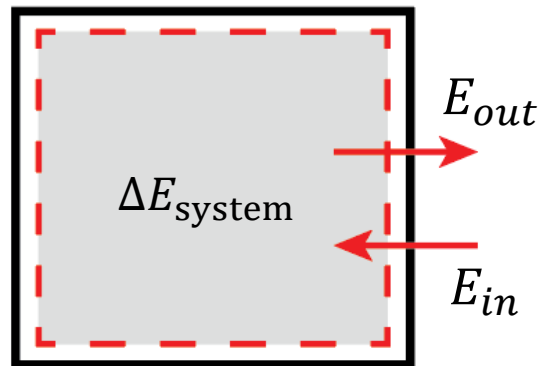
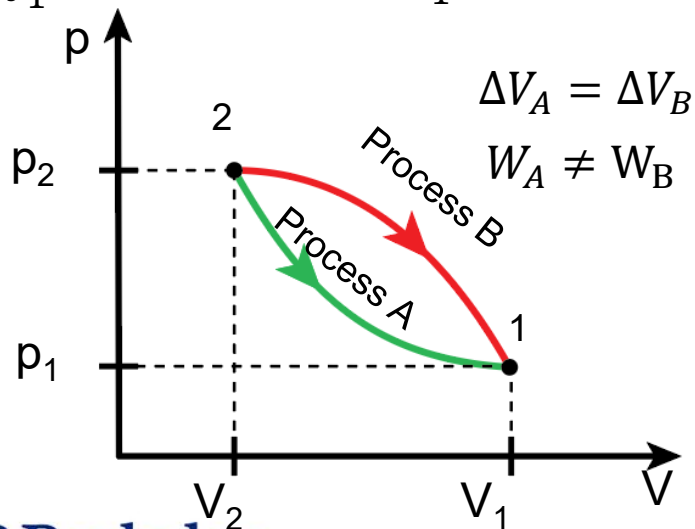
Inexact differentials $\delta Q; \delta W$

Exact differentials dV

Recap Lecture 2B

W and Q are path dependent

$$\int_1^2 dV = V_2 - V_1 \quad \int_1^2 \delta W = W_{12}$$



$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

$$\Delta U = Q - W$$

Energy transfer in/out of a closed system

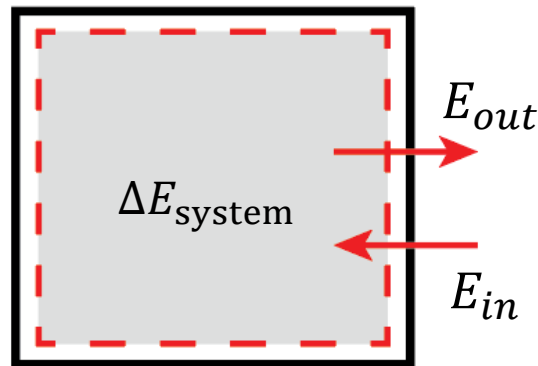
- Account for energy transfer with surroundings.

$$E_{in} - E_{out} = (Q_{in} - Q_{out}) + (W_{in} - W_{out})$$

Q $-W$

Heat in: $Q > 0$

Work out: $W > 0$



1st Law of thermodynamics: Example II

- Determine U_2 of the air inside the box.

- Define the system

$$\Delta U = Q - W$$

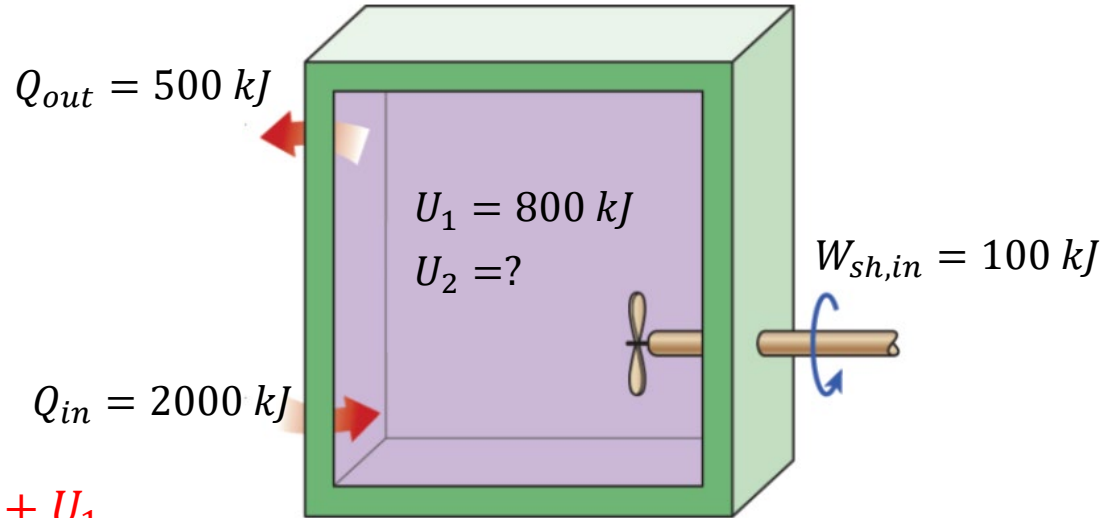
$$W = -100 \text{ kJ} ;$$

$$\Delta U = U_2 - U_1$$

$$Q = Q_{in} - Q_{out} = 1500 \text{ kJ} \quad Q_{in} = 2000 \text{ kJ}$$

$$U_2 = Q_{in} - Q_{out} + W_{shaft,in} + U_1$$

$$U_2 = Q_{in} - Q_{out} + W_{shaft,in} + U_1 = 2400 \text{ kJ}$$



Recap exercise: Isochoric process

Question:

1. What is the work extracted during a process of isochoric heating in a cylinder from state 1 to state 2:

To Do:

- Determine the relevant equation for Work.
- Simplify and rewrite that equation
- Go to pollev.com/claudiohail
- ... and answer question 1.

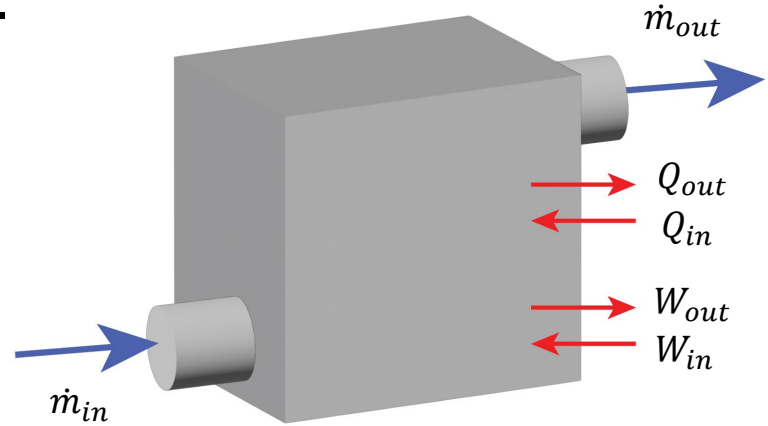


1st Law of thermodynamics: Open system

- Mass flows carry energy in/out.

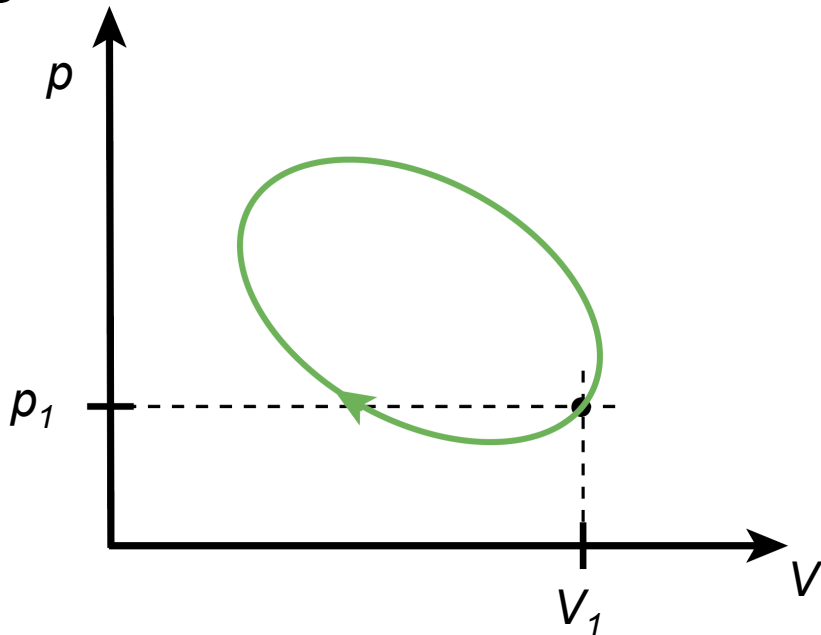
$$E_{in} - E_{out} =$$

$$\Delta E_{\text{system}} =$$



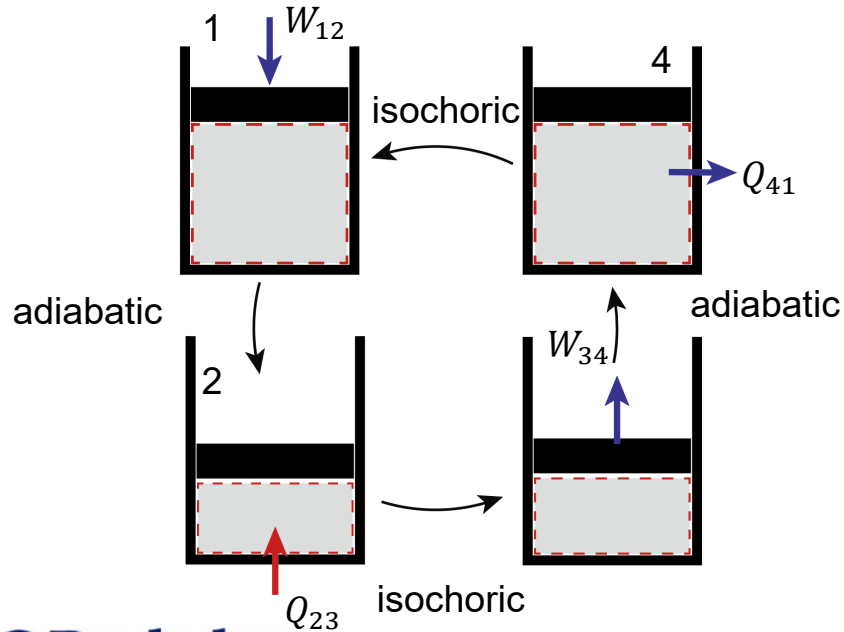
Energy analysis of thermodynamic cycles

- A closed system undergoing a thermodynamic cycle.



Energy analysis of thermodynamic cycles

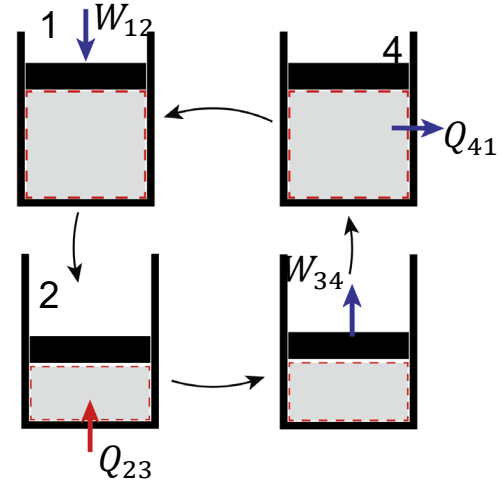
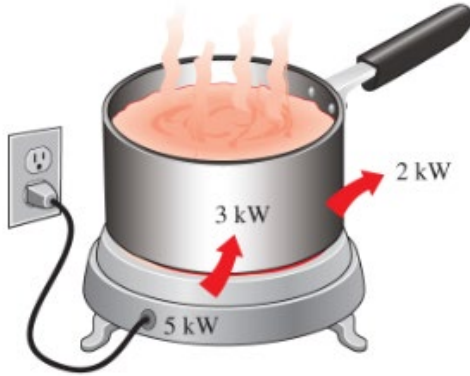
- Example of a closed cycle process of gas in a piston



Energy conversion efficiencies (I)

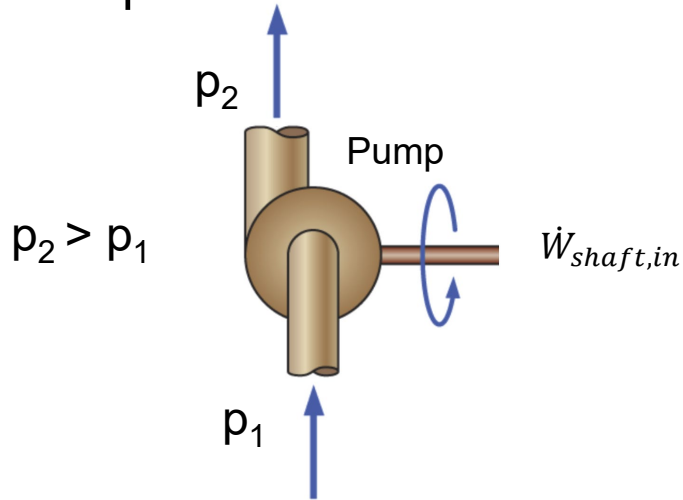
- Denoted by Greek letter η

$$\text{Efficiency} = \frac{\text{Desired output}}{\text{Required input}} \leq 1$$

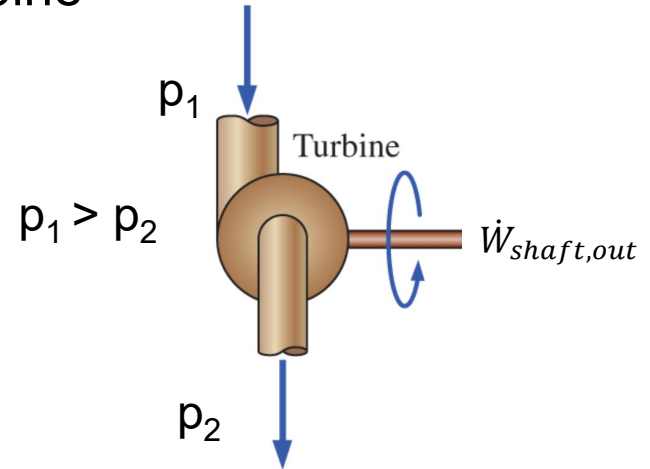


Energy conversion efficiencies (II)

- Pump

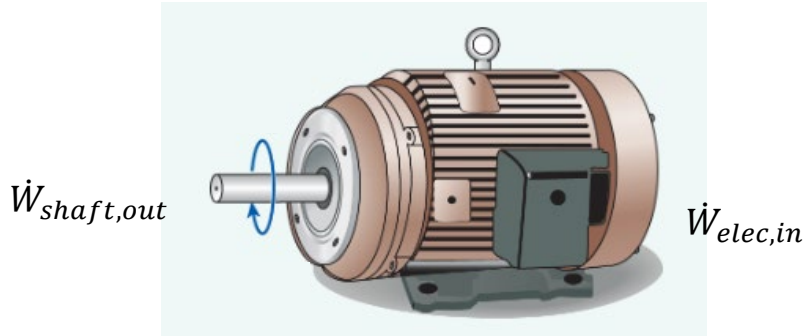


- Turbine

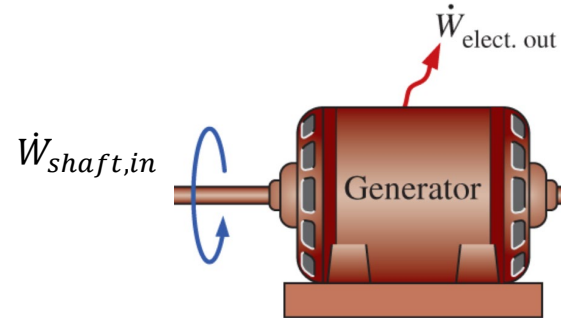


Energy conversion efficiencies (III)

- Motor

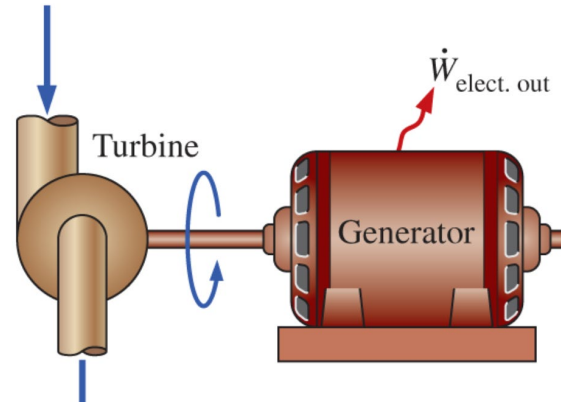


- Generator



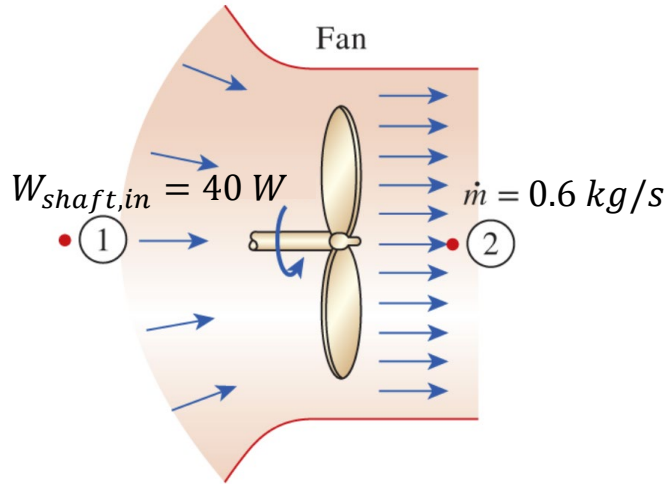
Energy conversion efficiencies

- Combining efficiencies of multiple devices in series:



Group Work: in groups of 2; solve the following problem.

Given:



$$V_1 = 0 ; V_2 = 10 \text{ m/s}$$

$$z_1 = z_2$$

$$P_1 \approx P_{atm} ; P_2 \approx P_{atm}$$

Question:

1. What is the efficiency of this device?
Give a formula and value.

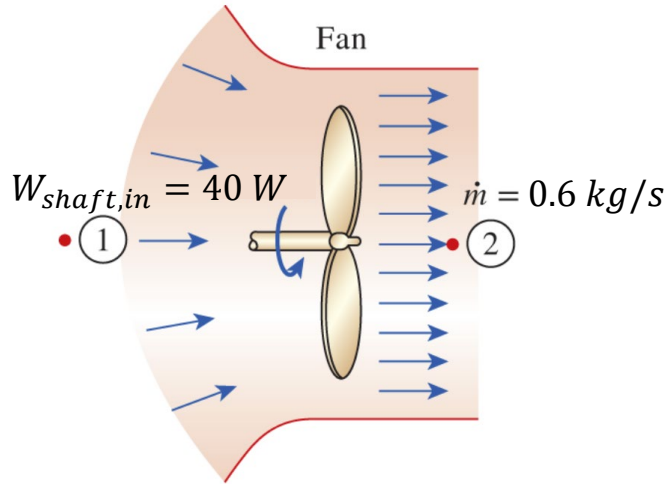
To Do:

- Define the thermodynamic system of interest.
- Determine the change in energy of the fluid.
- Determine the efficiency.
- Go to pollev.com/claudiohail
- ... and answer question 1.



Group Work: in groups of 2; solve the following problem.

Given:



$$V_1 = 0 ; V_2 = 10 \text{ m/s}$$

$$z_1 = z_2$$

$$P_1 \approx P_{atm} ; P_2 \approx P_{atm}$$

Summary: Energy & General Energy Analysis

Mass flow:

$$\dot{m} = \rho \dot{V} = \rho A_c V_{avg}$$

Mechanical energy:

$$\dot{E}_{mech} = \dot{m} e_{mech} = \dot{m} \left(\frac{P}{\rho} + \frac{V^2}{2} + gz \right)$$

Electrical work

$$W_e = UI \Delta t$$

Shaft work

$$W_{sh} = 2\pi n T$$

Spring work

$$W_{spring} = \frac{1}{2} k (x_2^2 - x_1^2)$$

Energy balance:

$$E_{in} - E_{out} = \Delta E_{system}$$

Energy balance (rate):

$$\dot{E}_{in} - \dot{E}_{out} = \frac{dE_{system}}{dt}$$

1st law closed system:

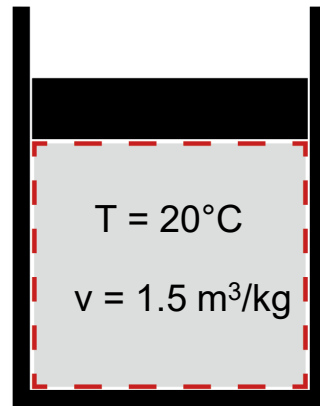
$$\Delta U = W - Q$$

Energy conversion efficiency:

$$Efficiency = \frac{\text{Desired output}}{\text{Required input}} \quad \eta = \frac{W_{net}}{Q_{in}}$$

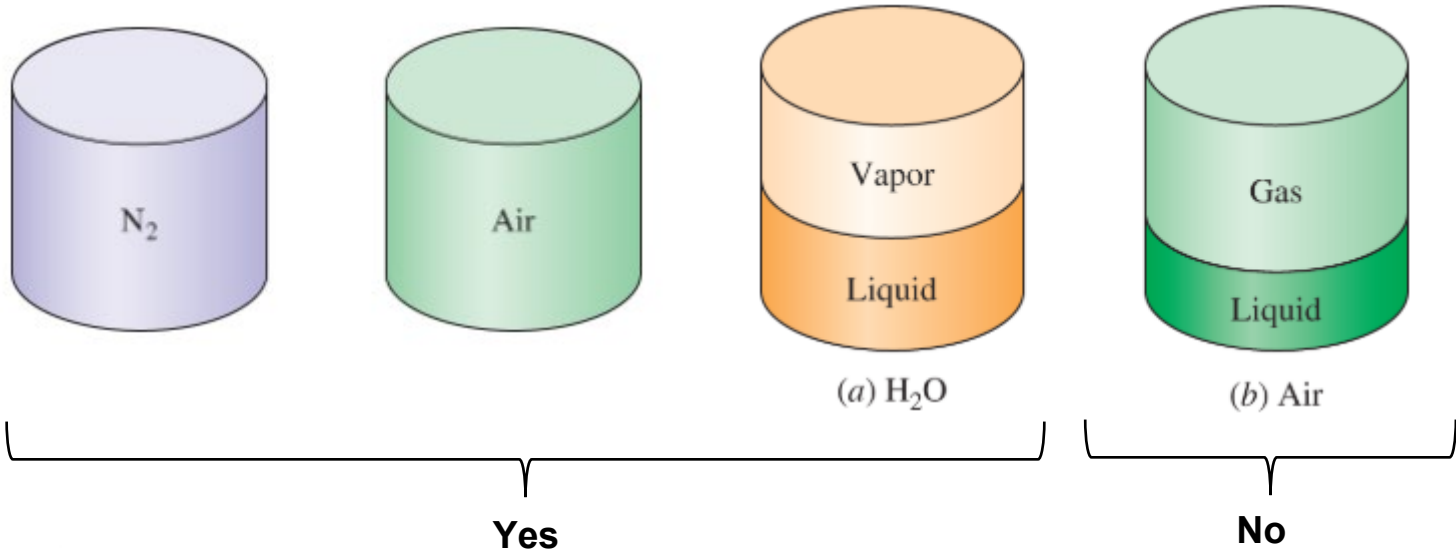
Properties of pure substances

- Thermodynamic description of the working fluids:
 - Water
 - Refrigerants (Ammonia, R134a, ...)
 - Gases – (often approximated as ideal gases)
 - Hydrocarbons
- Studying thermodynamic properties of different phases of matter.



Pure substances

- Definition: a fluid with a uniform chemical composition



Phases of matter

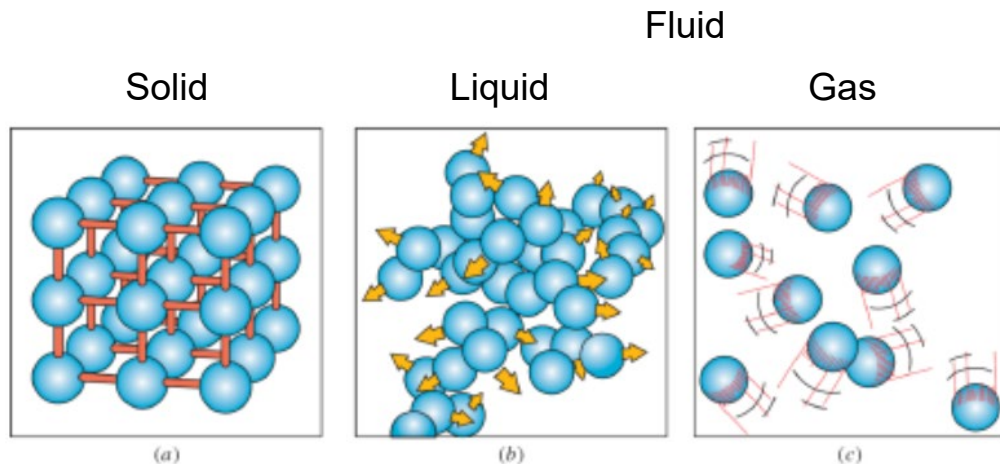


FIGURE 3–4

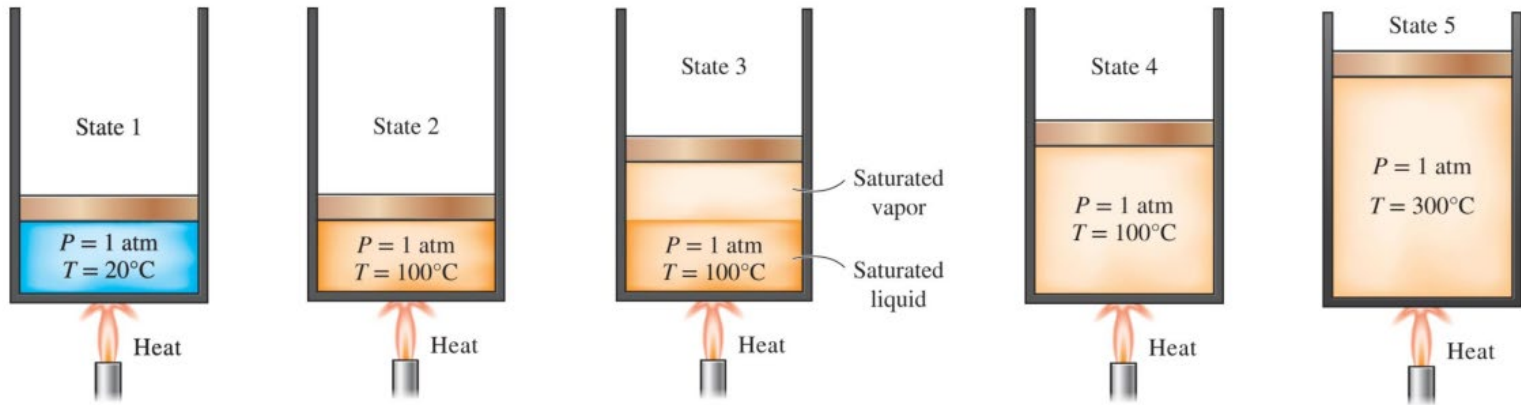
Melting
Freezing

Vaporizing
Condensation

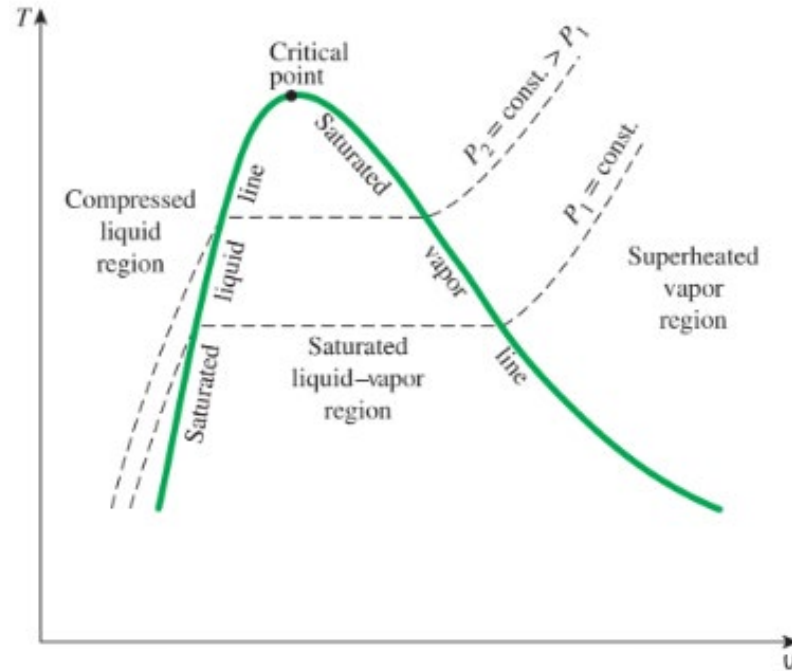
Sublimation
Deposition (resublimation)

Phase change processes of pure substances

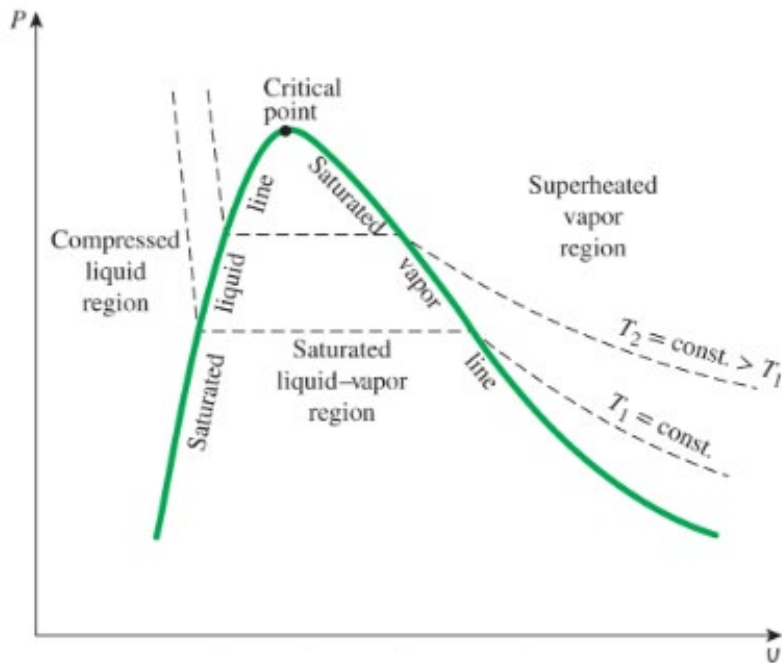
- Example: isobaric heating of Water



T-v diagram for pure substances



P-v diagram for pure substances



P-v diagram of a pure substance

Summary

- Understand energy analysis of open systems and cycles.
- Define energy conversion efficiency.
- Introduced pure substances and respective phase change processes.

Next up

- Thermodynamic property diagrams for pure substances.
- Procedures on determining the thermodynamic properties of pure substances.